

SCOPE OF WORK REPORT

Backward-Compatible FTIR
Calibration Using ML Version 4

Prepared By
Rowaina Reda
Salma Elfeel

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Objectives

- 1) Develop an accurate PLS regression model for prediction from NIR spectra.
- 2) Design a calibration set.
- 3) Ensure backward compatibility and robustness across devices of the same model type.
- 4) Explore the impact of sample design (holder type) on spectral stability.
- 5) Analyze device-induced spectral variability using PCA.

Phase 1: Literature Review and Planning

- Reviewed the fundamentals of FTIR spectroscopy, gaining an understanding of how infrared absorption patterns relate to molecular vibrations and how spectra are generated across different chemical compositions.
- Studied multivariate analysis techniques, especially Principal Component Analysis (PCA) and Partial Least Squares (PLS), to learn how these methods can reduce data dimensionality, extract key spectral features, and build predictive models for material concentration analysis.

Phase 2: Calibration Set Design

1. Generated multiple mixture 20 samples using Latin Hypercube Sampling (LHS) within the range of 0–10 grams for each **sample** to explore the experimental space systematically.
2. Observed that the reduced range resulted in fewer degrees of freedom, which limited dataset variability and did not decrease multicollinearity between variables.
3. Achieved an average inter-variable correlation of approximately 0.29, reflecting moderate correlation among mixture components.
4. Measured the grams of each ingredient for every sample using a sensitive balance with two-decimal precision to ensure accurate sample compositions.
5. Converted the measured gram values into percentage compositions.
6. Recorded the calculated percentages in a CSV file.

7. Sealed and thoroughly shook each sample in plastic bags to ensure homogeneity and uniform distribution of all components.
8. Transferred the homogenized samples into glass holders for long-term preservation and stability.

Phase 3: Spectral Data Acquisition

Applied FT-NIR spectroscopy to the samples, measuring those stored in plastic bags and glass holders.

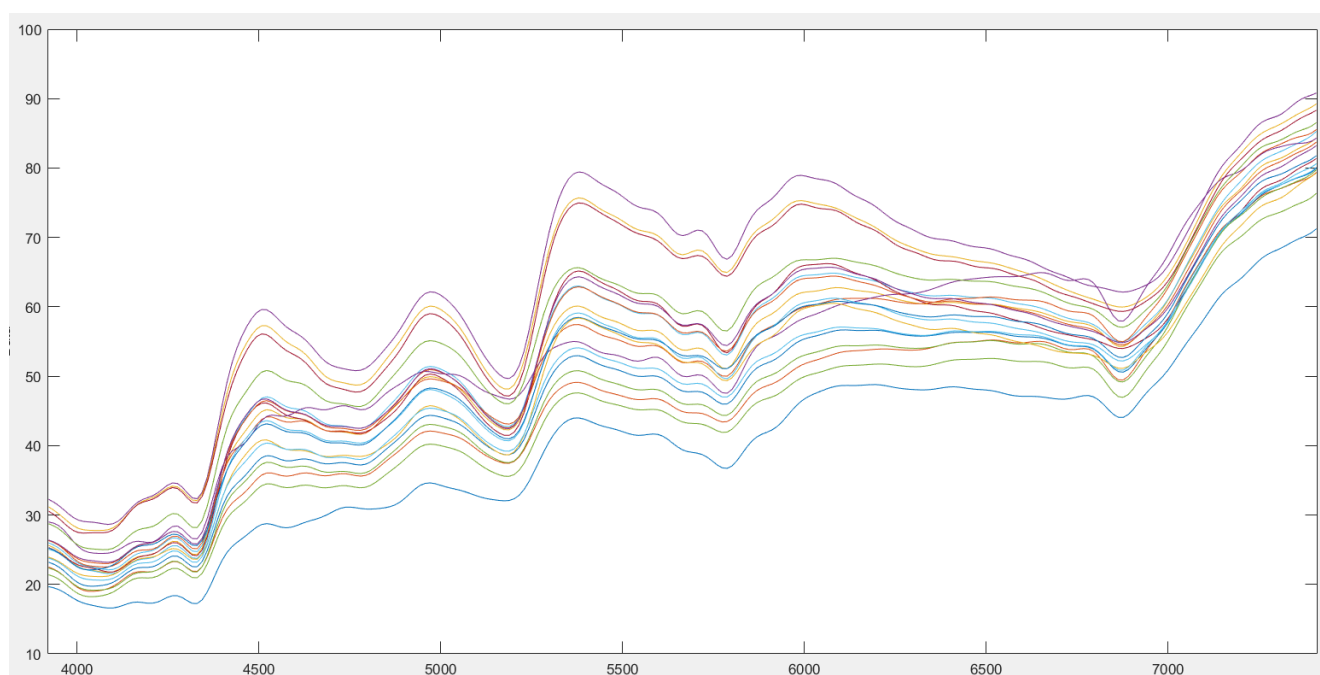


Figure 1: Spectra Data in Plastic Bags.

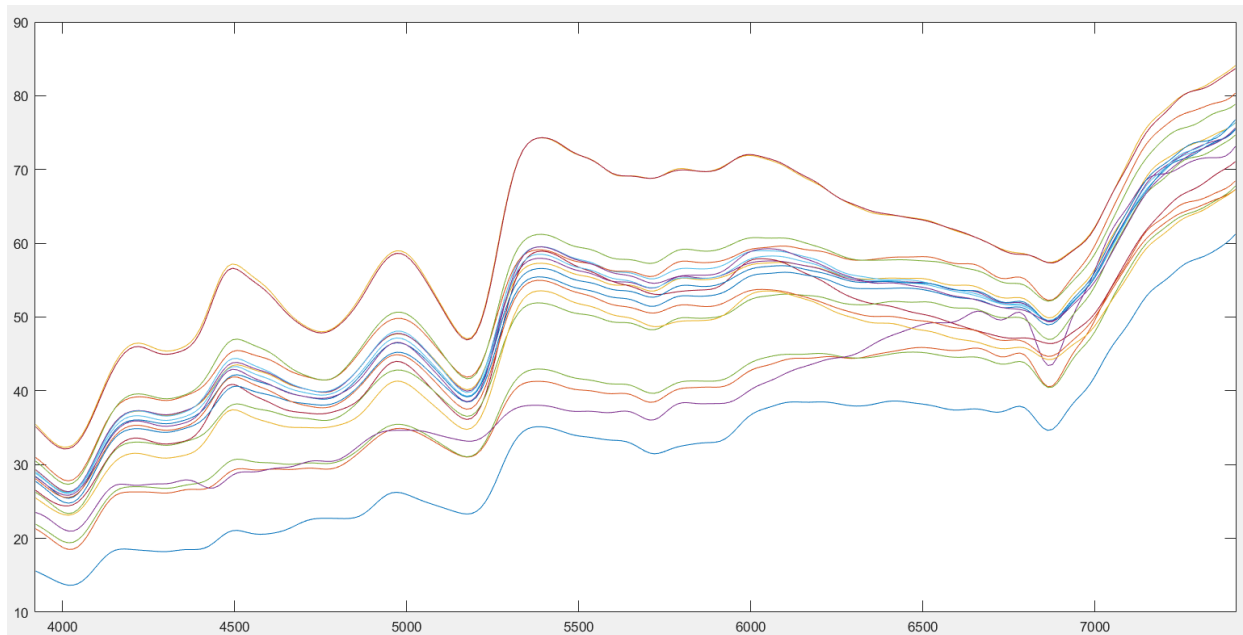


Figure 2: *Spectra Data in Glass Holders.*

Phase 4: Data Preprocessing

1. Explored multiple preprocessing options for the FT-NIR spectra to prepare the data for machine learning–based modeling.
2. Initially, the first derivative transformation was applied, which increased spectral complexity and reduced model generalizability, as it disrupted the linear relationship between spectra and sample compositions.
3. Revised the preprocessing strategy to preserve linearity and robustness, using the following steps:
 - Transmission-to-absorbance conversion
 - Standard Normal Variate (SNV) correction
 - Mean centering

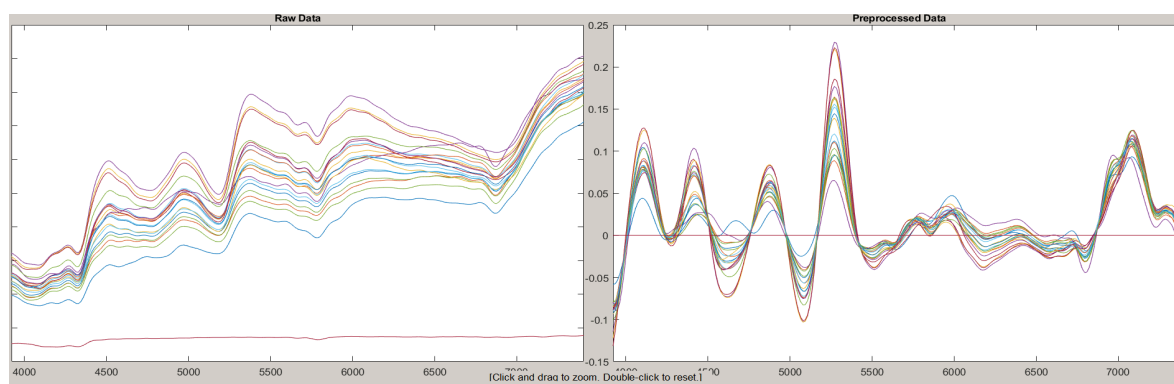
Phase 5: Model Development and Evaluation

Latent Variables over all models = 5

1. First Model (Plastic Samples, Bad Preprocessing)

Preprocessing: *Ratio to Reference* $\rightarrow$ *SNV* $\rightarrow$ *First derivative* (introduced complexity and poor generalization)

(*Ratio to Reference*) Taking the ratio to the reference yielded the red line in the figures above.

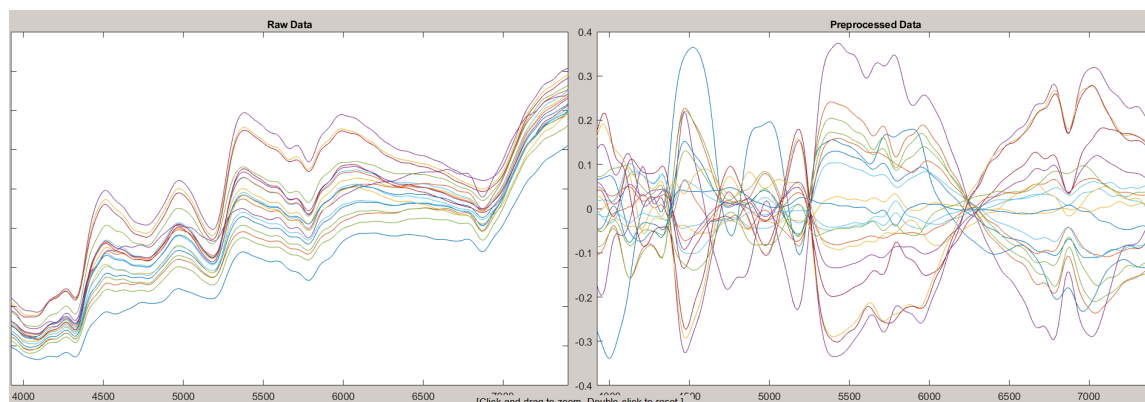


Results for Model R^2 Cross Validation:

<i>Starch (ST)</i>	<i>Protein (P)</i>	<i>Vitamin C (VC)</i>	<i>Na₂SO₄ (Salt)</i>
0.50	0.33	0.58	0.15

2. Second Model (Plastic Samples, Good Preprocessing)

Preprocessing: *Transmission to Absorbance* ($\log(1/T)$) $\rightarrow$ *SNV* $\rightarrow$ *Mean Centering*

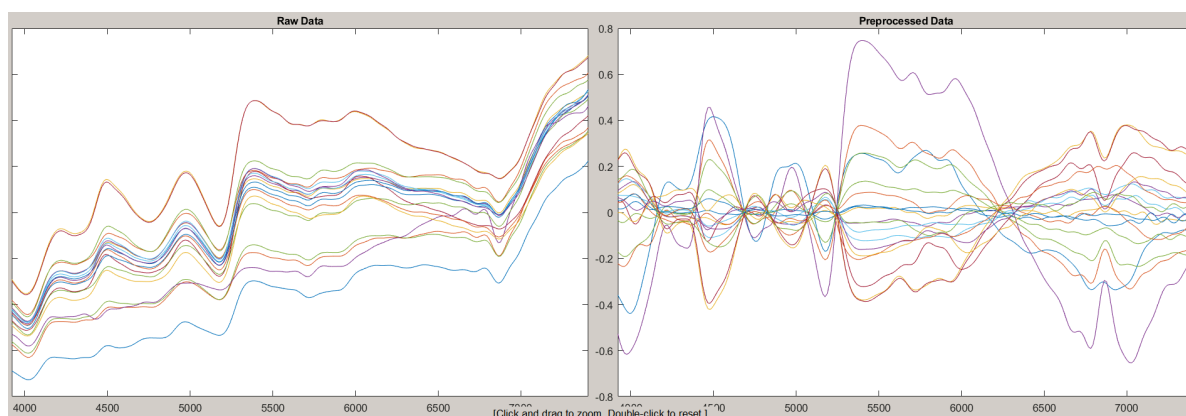


Results for Model R² Cross Validation:

<i>Starch (ST)</i>	<i>Protein (P)</i>	<i>Vitamin C (VC)</i>	<i>Na₂SO₄ (Salt)</i>
0.51	0.22	0.48	0.27

3. Third Model (Glass Samples, Good Preprocessing)

Preprocessing: *Transmission to Absorbance (log(1/T))* → *SNV* → *Mean Centering*



Results for Model R² Cross Validation:

<i>Starch (ST)</i>	<i>Protein (P)</i>	<i>Vitamin C (VC)</i>	<i>Na₂SO₄ (Salt)</i>
0.62	0.74	0.17	0.23

Model Observations

- Model 1** (plastic holders, first-derivative):
 Performed poorly due to increased spectral complexity and loss of linear relationship with ingredient concentrations.

- **Model 2** (plastic holders, same preprocessing):
Still underperformed; signal noise and multicollinearity affected the accuracy and generalization.
 - **Model 3** (glass holders):
Slight improvement in signal quality, but performance remained weak due to flawed preprocessing and mixture design.
-

Second Trial: Improved Sample Design and Modeling Pipeline

After the initial trial resulted in suboptimal model performance due to high multicollinearity and poor preprocessing choices, a second experimental setup was designed to overcome these issues and better reflect real-world mixture variation.

This trial introduced a new Latin Hypercube Sampling (LHS) approach with fewer constraints, better sample homogenization, and revised preprocessing and modeling techniques. The entire pipeline was optimized to increase model generalizability and ensure compatibility across different FTIR devices.

The following steps were followed carefully in the Second Trial:

1. Calibration Set Design

A new calibration set was generated using Latin Hypercube Sampling (LHS), but with a key modification:

- Each column sum was constrained to 10 grams to ensure realistic measurements for the balance while keeping each sample's total mass variable.
- This adjustment aimed to reduce multicollinearity between ingredients and increase the variability of samples without compromising physical feasibility.

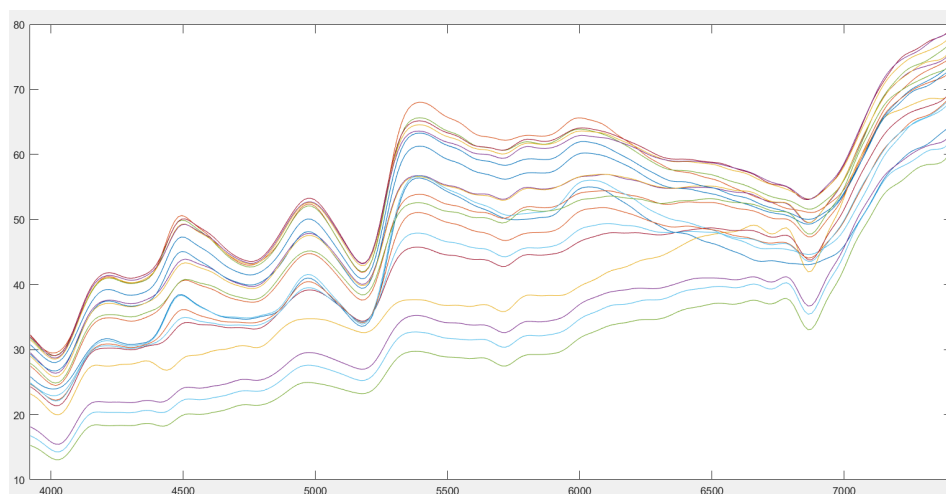


Figure 6: Spectra Data in Glass Bags trial two.

2. Model Development and Evaluation

Latent Variables over all models = 7, and the last recommended preprocessing.

○ Results for Modeling (Glass Samples):

<i>Starch</i>	<i>Protein</i>	<i>Vitamin C</i>	<i>Na₂SO₄</i>
0.78	0.73	0.69	0.11

The model is still underperforming for Salt.

○ Modeling with Oversampling

- i. Each of the 20 original samples was measured three times using FT-NIR, resulting in a total of 60 readings per spectrometer.

This helped expose the model to natural device-induced variability and improved its generalization without introducing synthetic data.

- ii. Cross-validation was performed at the sample level to prevent overfitting. The performance improvement suggests that this technique contributed to model stability and accuracy.
- iii. The data was split into 15 segments using the Venetian blinds method for cross-validation.

<i>Starch</i>	<i>Protein</i>	<i>Vitamin C</i>	<i>Na₂SO₄</i>
0.92	0.86	0.91	0.75

Observation: Oversampling improved the overall model performance, particularly for Na₂SO₄, resulting in a more balanced prediction across all components.

Final Model Selection and Backward Compatibility Testing

The final model was selected based on its performance with oversampled data and was applied to all remaining datasets from the other spectrometers. The model successfully demonstrated **backward compatibility**.

Cross-Spectrometer Evaluation (Same Device Type, Different Units)

<i>Devices</i>	<i>Starch (ST)</i>	<i>Protein (P)</i>	<i>Vitamin C (VC)</i>	<i>Na₂SO₄ (Salt)</i>
Test Set Device 1	0.92	0.86	0.92	0.68
Test Set Device 2	0.92	0.88	0.90	0.66
Test Set Device 3	0.93	0.87	0.94	0.64
Test Set Device 4	0.66	0.68	0.72	0.16

The results indicate good generalization and partial backward compatibility across FTIR devices of the same hardware type, with high R² values observed in three out of four test units. However, significantly lower performance on one device suggests that additional factors such as sample preparation or hardware variability may impact model consistency.

Model Evaluation (LODO Test)

To evaluate cross-device generalization, the model was trained using data from all spectrometers **except one**, which was reserved exclusively for testing. The **Leave-One-Device-Out (LODO)** strategy enabled the assessment of backward compatibility by simulating real-world variations.

The final model was developed using **Partial Least Squares (PLS)** regression with **9 components**, and preprocessing included **SNV** and **mean centering**. Evaluation metrics for the held-out spectrometer were as follows:

R^2	<i>Starch (ST)</i>	<i>Protein (P)</i>	<i>Vitamin C (VC)</i>	<i>Na₂SO₄ (Salt)</i>
Train 4 Devices	0.843	0.795	0.794	0.425
Test	0.914	0.835	0.899	0.644

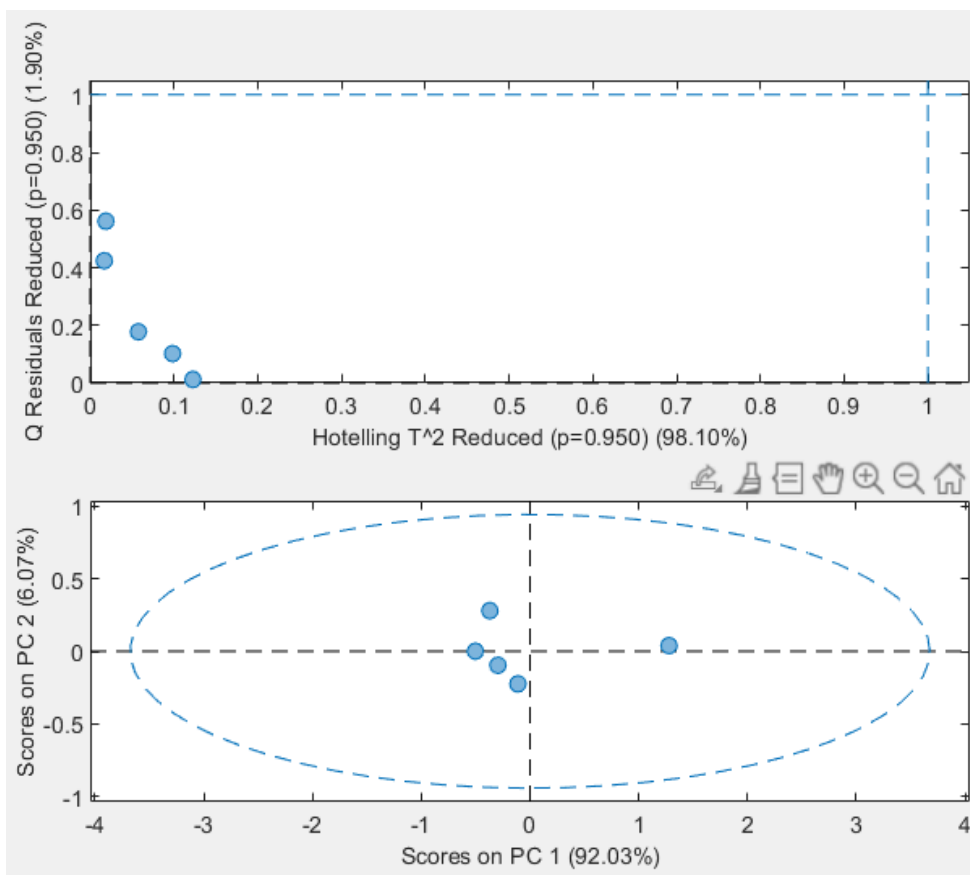
The LODO model showed slightly lower R^2 values compared to cross-validation within the same device, especially for vitamin C. However, this is expected, as LODO better reflects the model's ability to generalize across different instruments, which is critical for backward compatibility evaluation.

PCA Analysis Using Mean Spectra per Device

PCA was applied to the full set of spectral data (20 spectra per device) to investigate hardware-induced variability.

As shown in the score plot, most devices clustered closely, indicating similar spectral behavior. However, one device (Device 5) appeared as an outlier, showing a clear separation along PC1, which explains 92% of the total variance.

This suggests that Device 5 has a significantly different response compared to the others, which could affect model robustness and calibration transfer across instruments.



Proposed Future Improvements

1. Enhanced Model Validation

Given the limited dataset, adopting more robust validation strategies (e.g., repeated k-fold cross-validation or bootstrapping) could offer more reliable performance estimates.

2. Early Outlier Detection Using PCA

Applying PCA as an initial exploratory step can help detect hardware-induced shifts or sample anomalies before modeling, saving time and improving model stability.